Automated qPCR Analysis Report

Experiment: Auto qPCR run **Date:** 2025-11-20 **Analyst:** Auto-analyst

Summary of Results

- TargetX | Control_1: NA
- TargetX | Control_2: NA
- TargetX | ExpA_1: NA
- TargetX | ExpA_2: NA

QC Summary

Test	Status	Notes

NTC | Pass | {'TargetX': {'status': 'pass', 'notes': 'No amplification in NTC'}} | Replicate variance | OK | [{'sample': 'Control_1', 'target': 'GAPDH', 'status': 'warn', 'notes': 'insufficient replicates'}, {'sample': 'Control_1', 'target': 'TargetX', 'status': 'warn', 'notes': 'insufficient replicates'}, {'sample': 'Control_2', 'target': 'GAPDH', 'status': 'warn', 'notes': 'insufficient replicates'}, {'sample': 'Control_2', 'target': 'TargetX', 'status': 'warn', 'notes': 'insufficient replicates'}, {'sample': 'ExpA_1', 'target': 'GAPDH', 'status': 'warn', 'notes': 'insufficient replicates'}] | Housekeeping | OK | {'GAPDH': {'status': 'fail', 'sd': 0.3960744879438714}, 'ACTB': {'status': 'missing', 'notes': 'No housekeeping measurements'}} |

Cleaned CT Table

Sample	Target	CT_mean	n_rep	CT_sd
Control_1	GAPDH	18.4	1	nan
Control_1	TargetX	22	1	nan
Control_2	GAPDH	18.6	1	nan
Control_2	TargetX	22.5	1	nan
ExpA_1	GAPDH	19.1	1	nan
ExpA_1	TargetX	19.3	1	nan
ExpA_2	GAPDH	19.4	1	nan
ExpA_2	TargetX	19.7	1	nan
NTC	TargetX	nan	0	nan

ΔCT / ΔΔCT Analysis

Sample	Target	DeltaCT	DeltaDeltaCT	FoldChange
Control_1	TargetX	3.6	nan	nan
Control_2	TargetX	3.9	nan	nan
ExpA_1	TargetX	0.2	nan	nan
ExpA_2	TargetX	0.3	nan	nan

Plots

Interpretation

Auto-generated interpretation - review carefully.